Hybrid Cloud Detection in Multispectral Satellite Imagery using Spectral Thresholding and CNN Segmentation

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Abstract— Cloud detection is a critical preprocessing step in satellite image analysis, as cloud coverage can significantly affect the accuracy of remote sensing applications including agricultural monitoring, land-use analysis, environmental observation, and disaster management. This paper presents a hybrid cloud detection framework using Landsat-9 multispectral satellite imagery by integrating spectral thresholding techniques with convolutional neural network (CNN)-based segmentation.

Initially, cloud regions are identified using spectral characteristics derived from Red, Near Infrared (NIR), and Thermal Infrared (TIR) bands. The generated cloud masks are further refined using a lightweight U-Net inspired CNN segmentation model trained with QA_PIXEL reference cloud masks. Spectral analysis, threshold-based cloud masking, and deep learning-based segmentation were implemented and evaluated using Intersection over Union (IoU) metrics.

Experimental results demonstrate that the CNN-based segmentation approach significantly improves cloud-region representation and spatial overlap compared to traditional spectral thresholding methods. The proposed framework highlights the effectiveness of combining rule-based spectral analysis and deep learning techniques for reliable automated cloud detection in modern remote sensing workflows.

Keywords— Cloud Detection, Remote Sensing, CNN Segmentation, Landsat-9, Multispectral Imagery, Deep Learning, Image Processing

I. INTRODUCTION

Clouds routinely complicate remote-sensing work by obscuring features on the ground. Any field that relies on satellite data—whether environmental monitoring, crop assessment, or urban studies—benefits from accurate cloud masking. This work focuses on the development and evaluation of automated cloud-detection methods using publicly available multispectral satellite imagery and Python-based image-processing techniques.

With the increasing availability of high-resolution satellite imagery, automated cloud detection has become more important than ever. Manual inspection is no longer feasible due to the large volume of data generated daily. Hence, efficient algorithms are required to ensure accurate preprocessing before further analysis such as classification, object detection, and environmental monitoring.

II. LITERATURE REVIEW

Cloud detection has been widely studied in remote sensing due to its critical role in improving image quality. Traditional methods mainly rely on spectral thresholding techniques, where cloud pixels are identified based on their reflectance and temperature properties. These methods are simple and computationally efficient but often fail in complex scenarios such as thin clouds or bright surfaces like snow and sand.

Recent advancements have introduced machine learning and deep learning approaches for cloud detection. Convolutional Neural Networks (CNNs) have shown significant improvements by learning spatial and spectral features automatically. U-Net architectures, in particular, are widely used for pixel-wise segmentation tasks.

Studies such as Li et al. (2019) and CMIX (2022) demonstrate that deep learning models outperform classical approaches in terms of accuracy and robustness. However, they require large datasets and higher computational power. This project combines both approaches to leverage their advantages.

III. SYSTEM OVERVIEW

A. Simple Input Requirements

The user only needs to provide the required satellite image bands used in the project:

- Red band
- NIR (Near Infrared) band
- Thermal Infrared (TIR) band

Satellite Image Analysis Cloud …

These are easily available from public datasets like Landsat-9 and 38-Cloud.

B. Easy Workflow

The overall workflow is straightforward:

1. Load the satellite bands
2. Normalize / preprocess
3. Run cloud detection using:
   - Spectral Thresholding
   - CNN Segmentation
4. Generate a binary cloud mask output (Cloud = 1, Clear = 0)

C. User-Friendly Cloud Mask Output

The output is very easy to interpret:

- White pixels → Cloud
- Black pixels → Clear land

Also, the project includes overlay visualization, which makes it easier to verify results visually.

IV. METHODOLOGY

The methodology followed in this project includes acquiring multispectral satellite images, preprocessing the data, applying cloud detection algorithms, and evaluating the results using standard performance metrics. The complete workflow is designed to compare a traditional spectral thresholding approach with a machine learning based CNN segmentation approach.

The methodology follows a hybrid approach combining rule-based and learning-based techniques. Initially, spectral thresholding provides a baseline detection using physical properties of clouds. This is followed by CNN-based segmentation, which enhances detection by capturing spatial patterns and contextual information. The comparison between these two methods helps in understanding the trade-off between simplicity and accuracy.

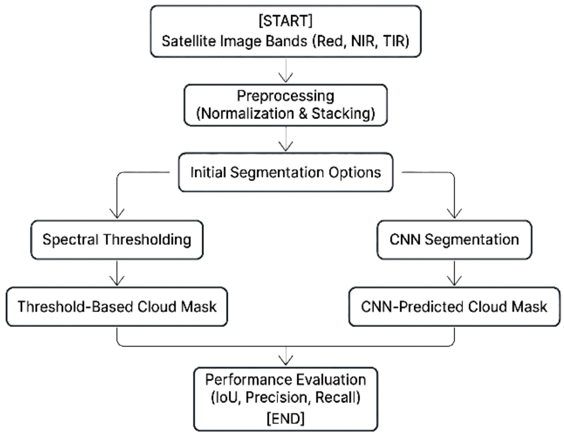

Fig. 1. Overall workflow of the proposed cloud detection system.

A. Data Acquisition

The satellite imagery used in this project was obtained from the Landsat-9 Level-2 Surface Reflectance dataset provided by the United States Geological Survey (USGS). The selected multispectral bands included Red (SR_B4), Near Infrared (SR_B5), Thermal Infrared (ST_B10), and QA_PIXEL bands. The QA_PIXEL layer was utilized as the reference ground-truth cloud mask for threshold validation and CNN-based segmentation training.

The experiments were conducted using Landsat-9 Level-2 multispectral satellite imagery consisting of Red (SR_B4), Near Infrared (SR_B5), Thermal Infrared (ST_B10), and QA_PIXEL bands. The QA_PIXEL layer provided reference cloud-mask labels used for evaluation and CNN-based segmentation training.

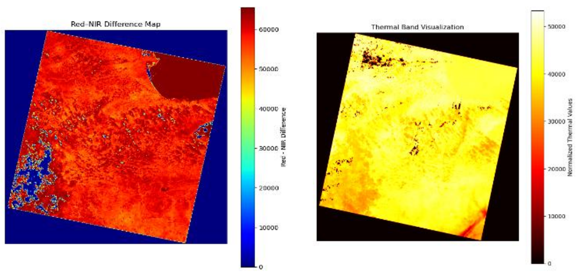

Fig. 2. Spectral analysis using Red–NIR difference and normalized thermal band for identifying cloud regions.

B. Preprocessing

Before applying cloud detection, preprocessing is performed to improve the quality of the input data. The preprocessing steps include normalization of pixel values across different bands, resizing or cropping images to a required region of interest, and stacking the selected bands into a single multispectral input image. If needed, basic radiometric and geometric corrections are also applied. This step ensures that all bands are aligned properly and are suitable for thresholding and CNN training. All spectral bands were normalized to a common range before feature extraction and segmentation.

C. *Cloud Detection using Spectral Thresholding*

In the first approach, cloud detection is performed using a rule-based spectral thresholding method. A Red–NIR difference map is computed to highlight reflective cloud regions. Additionally, the thermal band is used to detect colder regions, since clouds usually appear cooler than the land surface. A pixel is classified as cloud if the Red–NIR difference exceeds a fixed threshold and the thermal temperature value is below a fixed temperature limit. The output of the spectral thresholding process is a binary cloud mask, where cloud pixels are represented by 1 and non-cloud pixels are represented by 0.

- The spectral difference between the Red and Near Infrared (NIR) bands was calculated using:

$$\boxed{D = Red - NIR}$$

where (R) represents the Red band reflectance and (NIR) represents the Near Infrared band reflectance.

- Cloud pixels were identified using spectral thresholding conditions defined as:

$$\boxed{M(x, y) = \begin{cases} 1, \text{if } R(x, y) > Tr \text{ and } D(x, y) > Td \\ 0, \text{Otherwise} \end{cases}}$$

where (M (x, y)) represents the binary cloud mask, (Tr) is the Red-band threshold, and (Td) is the spectral difference threshold.

D. *Cloud Detection using CNN Segmentation*

In the second approach, a convolutional neural network (CNN) model is trained to perform pixel-wise cloud segmentation. The input to the CNN consists of stacked multispectral bands, and the output is a predicted cloud mask. The model is trained using image-mask pairs from the dataset. A lightweight U-Net inspired convolutional neural network architecture was implemented for pixel-wise cloud segmentation. The trained CNN model automatically learns spatial and spectral representations from multispectral satellite imagery, resulting in improved cloud-region segmentation performance. The CNN model was trained using Binary Cross entropy loss defined as:

$$\boxed{L = -\frac{1}{N}\sum_{i=1}^{N}[yilog(y\hat{\ }i) + (1 - yi)log(1 - y\hat{\ }i)]}$$

where (yi) represents the ground-truth label and (y^i) represents the predicted probability.

E. *Evaluation and Performance Metrics*

To evaluate the performance of both methods, standard metrics are used. Intersection over Union (IoU) is used as the primary metric to measure overlap between predicted masks and ground-truth masks. Precision and Recall are also calculated to analyze the correctness of cloud detection. In addition to numerical evaluation, visual inspection is performed by overlaying the predicted cloud masks on the original satellite images. This helps verify the quality of segmentation results in practical scenarios.

- The segmentation performance was evaluated using Intersection over Union (IoU), calculated as:

$$\boxed{IoU = \frac{TP}{TP + FP + FN}}$$

where (TP), (FP), and (FN) denote true positives, false positives, and false negatives, respectively.

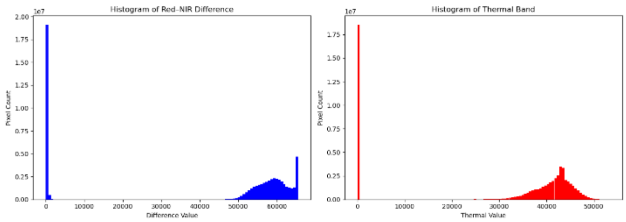

Fig. 3. Histogram distribution of Red–NIR values and normalized thermal responses used for threshold-based cloud detection.

F. *Experimental Setup*

The proposed cloud detection and segmentation framework was implemented using Python in a Jupyter Notebook environment. Landsat-9 Level-2 multispectral satellite imagery was utilized for experimental analysis. The input data consisted of Red (SR_B4), Near Infrared (SR_B5), Thermal Infrared (ST_B10), and QA_PIXEL bands obtained from the United States Geological Survey (USGS) satellite data repository.

Initially, spectral thresholding techniques were applied using Red-band reflectance and Red–NIR spectral difference analysis to generate baseline cloud masks. The QA_PIXEL band was used as the reference ground-truth cloud mask for evaluation and CNN training purposes.

For deep learning-based cloud segmentation, a lightweight U-Net inspired convolutional neural network architecture was implemented using TensorFlow and Keras libraries. The input multispectral images were resized to (256 \times 256) resolution for computational efficiency. Data augmentation techniques including horizontal flipping, vertical flipping, and image rotation were applied to improve training diversity.

The CNN model was trained using the Adam optimizer and Binary Cross entropy loss function. Segmentation performance was quantitatively evaluated using the Intersection over Union (IoU) metric to compare the overlap between predicted cloud masks and QA_PIXEL reference masks.

Parameter	Value
Programming Language	Python
Development Environment	Jupyter Notebook
Libraries Used	TensorFlow, Rasterio, NumPy, OpenCV, Matplotlib, Scikit-learn
Satellite Data Source	Landsat-9 Level-2 Surface Reflectance Data
Input Bands	Red (SR_B4), NIR (SR_B5), Thermal (ST_B10)
Ground Truth Mask	QA_PIXEL Cloud Mask
Detection Methods	Spectral Thresholding and CNN Segmentation
CNN Architecture	Lightweight U-Net Inspired CNN
Optimizer	Adam
Loss Function	Binary Crossentropy
Image Resolution	256 × 256
Evaluation Metric	Intersection over Union (IoU)
Thresholding IoU	0.1649
CNN Segmentation IoU	0.4762
Output	Binary Cloud Segmentation Mask

Table 1. Experimental Configuration

V. RESULTS

The spectral thresholding approach produced partial overlap with the QA_PIXEL reference cloud masks, achieving an Intersection over Union (IoU) score of approximately 16.5%. Although the threshold-based method successfully identified dense cloud regions, it struggled to capture fragmented cloud structures and thin cloud formations across the satellite scene.

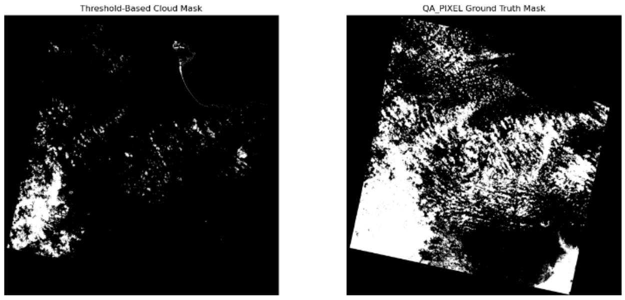

Fig. 4. Comparison between spectral thresholding output and QA_PIXEL reference cloud mask used for baseline cloud detection evaluation.

To improve segmentation performance, a lightweight U-Net inspired CNN model was trained using multispectral satellite bands and QA_PIXEL cloud masks. The CNN-based segmentation approach achieved an IoU score of approximately 47.6%, significantly outperforming the traditional thresholding method. Visual inspection of the segmentation outputs indicated improved spatial continuity and cloud-region overlap compared to spectral thresholding methods.

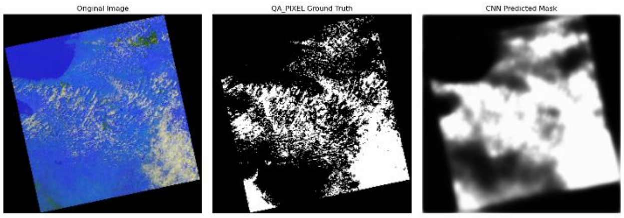

Fig. 5. Comparison between the original multispectral satellite image, QA_PIXEL reference cloud mask, and CNN-predicted cloud segmentation output.

The experimental results demonstrate that deep learning-based segmentation methods provide more robust cloud detection performance than rule-based spectral thresholding techniques, particularly for complex cloud distributions and spatially varying atmospheric patterns.

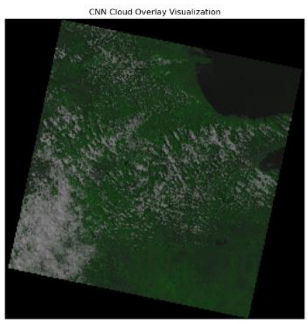

Fig. 6. Cloud detection overlay generated on the red spectral band after applying thresholding and segmentation methods.

VI. APPLICATIONS

The proposed cloud detection and segmentation framework can be applied in several remote sensing and environmental monitoring applications. Accurate cloud detection is important for improving the quality of satellite-based Earth observation systems and minimizing cloud-related interference in image analysis workflows.

Potential applications include:

- Weather monitoring and cloud movement analysis
- Satellite image preprocessing for remote sensing applications
- Agricultural and vegetation monitoring
- Disaster management and environmental surveillance
- Climate and atmospheric studies

The integration of spectral thresholding and CNN-based segmentation techniques provides a scalable approach for automated cloud detection in multispectral satellite imagery.

VII. CONCLUSION

This paper presented a hybrid cloud detection framework combining spectral thresholding techniques and CNN-based segmentation for multispectral satellite imagery analysis. Landsat-9 Level-2 satellite data consisting of Red, Near Infrared, Thermal Infrared, and QA_PIXEL bands were utilized to perform cloud-region extraction and segmentation experiments.

Initially, spectral thresholding methods based on Red-band reflectance and Red–NIR spectral analysis were implemented to generate baseline cloud masks. Although the thresholding approach successfully identified major dense cloud regions, it showed limitations in detecting fragmented cloud structures and spatially complex cloud formations.

To improve segmentation performance, a lightweight U-Net inspired convolutional neural network was developed using multispectral satellite inputs and QA_PIXEL reference masks. Experimental evaluation using Intersection over Union (IoU) metrics demonstrated that the CNN-based segmentation approach significantly outperformed traditional thresholding techniques by learning spatial cloud features and generating smoother cloud-region representations.

The experimental results demonstrate the effectiveness of integrating spectral analysis and deep learning techniques for automated cloud detection in multispectral remote sensing applications. Future work may include training on larger satellite datasets, improving segmentation accuracy using advanced deep learning architectures, and extending the framework for real-time satellite image processing systems.

VIII. REFERENCES

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